PHASE 1 PLAN

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AURA TEAM

11/13/25

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ANALYSIS OF THE 'PHASE 1 PLAN

INTRODUCTION

This plan is based on three main pillars. These pillars will guide our learning, our project work, and our professional growth. The goal is to build strong basics in programming, manage our university studies, and create a good online presence.

PILLAR 1: CORE OOP

We will work on OOP for four weeks. This project will help us use and understand all the OOP concepts we learn. Working through a project is one of the best ways to learn programming in a simple and clear way.

PILLAR 2: UNIVERSITY STUDIES

We will give two days of the week only to university study OR three days, we will manage university study and project work together.
We will also keep preparing for exams so our project does not stop during exam time.

PILLAR 3: PROFESSIONAL BRANDING

We will use GitHub, LinkedIn and Medium to build a strong professional profile. These platforms will help us show our work, explain our progress, and present our skills in a simple and clear way. This will support us when we apply for internships or future opportunities.

WEEK CYCLE	
Pillars	**Working Days per Week**
1	4 days
2	2 days
3	6 days
Sunday	1 day (free Day) No Work

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TARGET OF PHASE 1

1. OOPS Complete With Project.
2. University Studies
3. Professional Branding (The "Showcase")

DURATION PHASE 1

- 28 Days

OOPS FOUR WEEK PLAN

TARGET

Master Object-Oriented Programming (OOP) concepts in 28 days with both theoretical understanding for university exams and practical application for real-life projects.

CHAPTER 1 —OOP FOUNDATIONS

- What is OOP
- Why OOP evolved
- Difference between OOP vs Procedural
- Principles: abstraction, encapsulation, inheritance, polymorphism

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CHAPTER 2 — CLASSES & OBJECTS

- Class structure
- Fields, methods
- Creating objects
- Object state
- Encapsulation (private fields + getters/setters)
- Package access

CHAPTER 3 — CONSTRUCTORS

- Default constructor
- Parameterized constructor
- Overloading
- Copy constructor
- Memory allocation for objects
- Object as reference

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CHAPTER 4 — STATIC & BASIC FEATURES

- Static fields/methods
- finalize()
- Garbage collection
- Immutable objects
- Wrapper classes

CHAPTER 5 — UML & RELATIONS

- Class diagrams
- Associations
- Aggregation vs Composition
- Forward & reverse engineering

CHAPTER 6 — INHERITANCE

- Superclasses/subclasses
- super keyword
- protected access
- final keyword
- Constructor chaining
- Overriding vs overloading

CHAPTER 7 — POLYMORPHISM

- Runtime polymorphism
- Dynamic binding
- Upcasting/downcasting
- equals(), toString()
- Subtyping relations

CHAPTER 8 — ABSTRACT & INTERFACES

- Abstract classes
- Abstract methods
- Interfaces
- Comparable & Cloneable

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CHAPTER 9 — COLLECTIONS & GENERICS

- ArrayList
- Generic classes
- Type safety
- Static vs dynamic typing

CHAPTER 10 — FILE I/O

- Reading text files
- Writing text files
- Binary streams
- Object serialization

CHAPTER 11 — GUI COMPONENTS

- Label, Button, TextField
- TextArea, ScrollBar
- ComboBox, RadioButton

CHAPTER 12 — LAYOUT MANAGERS

- BorderLayout
- FlowLayout
- GridLayout

CHAPTER 13 — EVENT HANDLING

- Event-driven programming
- Event types
- Listener interfaces
- Inner classes
- Anonymous classes
- Lambda event handling
- Window listeners
- Adapter classes

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OOPS WEEKS PLAN		
Week	**Units**	**Goal**
1	1,2,3,4,5,6,7	Build a solid foundation in OOP concepts.
2	8,9,10	Apply intermediate OOP features and relationships.
3	11,12,13	Master advanced OOP concepts and prepare for real-world applications
4	Projects	Apply OOP concepts to full-scale real-life projects.

ONLINE PROFESSIONAL PROFILE AND SHOWCASE

Target

- Build a strong online professional profile and actively showcase our skills, projects, and expertise across platforms to establish a credible digital presence.

WORKING		
DAYS	**TOPIC**	**Goal**
1	LinkedIn	Build a Professional and optimized Profile.
2	GitHub	Create an Account Professional and optimized Profile
3	Medium	Create an Account Professional and optimized Profile

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Platform	Frequency	What to Post (The "Job")
GitHub	**3-5 time/Week**	**Do the work.** Commit meaningful progress on your 1-2 key projects.
LinkedIn	**3-5 time/Week**	**Show the work.** Post about your project, share your knowledge, and engage with others.
Medium	**1-2 times/month**	**Explain the work.** Write a high-quality article that tells the story behind your GitHub project.

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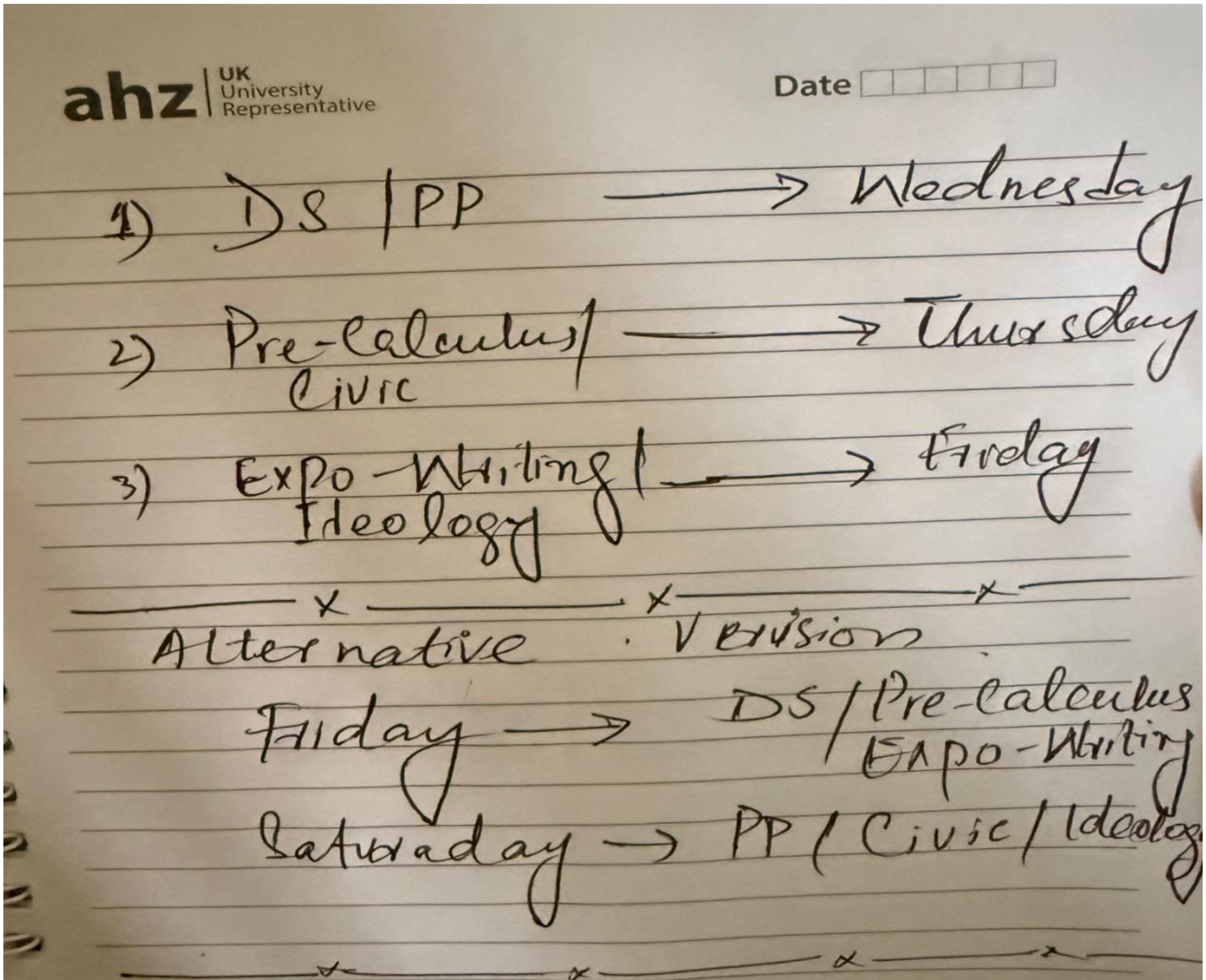
ahz | UK University Representative

Date

1) DS | PP → Wednesday

2) Pre-Calculus / Civic → Thursday

3) Expo-Writing / Ideology → Friday

Alternative . Revision

Friday → DS / Pre-Calculus / Expo-Writing

Saturday → PP / Civic / Ideology

REWARDS DAY

- There will Reward Day after Completing Phase 1
- Best Performer Will get the Reward

BEST LUCK

THE END